

Linear Algebra & Differential Equations for Quants

Mathematical Intuition for Quantitative Finance
Geometry, Dynamics, and Stability Made Visual

Amit Kumar Jha, Quant Analyst at UBS

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The 3-Second Answer (Module 8)

Question: "Why can't we solve Heston PDE in closed form?"

Your opening line:

"Stochastic vol makes coefficients random. The PDE becomes $\frac{\partial V}{\partial t} + vS^2\Gamma + \dots$ where v is stochastic. No constant coefficients no symmetry need Fourier transform or MC. The operator is non-elliptic."

Why this wins:

- Identifies random coefficients
- Links to symmetry loss
- Mentions solution alternatives

Module 9: When PDEs Break

Module Overview

What this module is about:

PDEs fail when dimensionality is high, boundaries are free, or operators are non-local.

Why this matters:

Most exotic products need numerical methods PDE is too slow or impossible.

After this module: You will know when to abandon PDE and use MC.

9.1 The Curse of Dimensionality

BS: 2 dimensions (S, t) fine

Basket: 5 underlyings 5 spatial dimensions grid size = N^5

Core Formula

$$\text{Grid points} = N^{\text{dimensions}}$$

If $N = 100$: - 1D: 100 points - 2D: 10,000 points - 5D: 10 billion points (infeasible)

"5 Dimensions = MC Only"

PDE solvers scale exponentially. MC scales linearly (just add more paths). For >3 dimensions, MC wins.

9.2 Free Boundaries (American Options)

American option PDE:

$$\min \left(\frac{\partial V}{\partial t} + \mathcal{L}V, V - \text{Payoff} \right) = 0$$

Problem: Boundary $S^*(t)$ is unknown. Must solve PDE + variational inequality.

Practitioner Insight

American Options: PDE vs MC:

PDE approach: - Use finite difference with PSOR (Projected SOR) - Iterate: solve PDE, check exercise, project to payoff - Slow but accurate for low dimensions

MC approach: - Longstaff-Schwartz regression - Approximate continuation value $C(S, t) \approx \beta\phi(S)$ - Fast but biased by regression error

Desk choice: - Vanilla American: PDE (1D) - Bermudan on basket: MC (3D+)

Real-Time Detection: Monitor $\kappa = \lambda_1 / \lambda_2$ intraday. When $\kappa > 8$, cut positions *before* risk limits trigger.

The Correlation Phase Transition

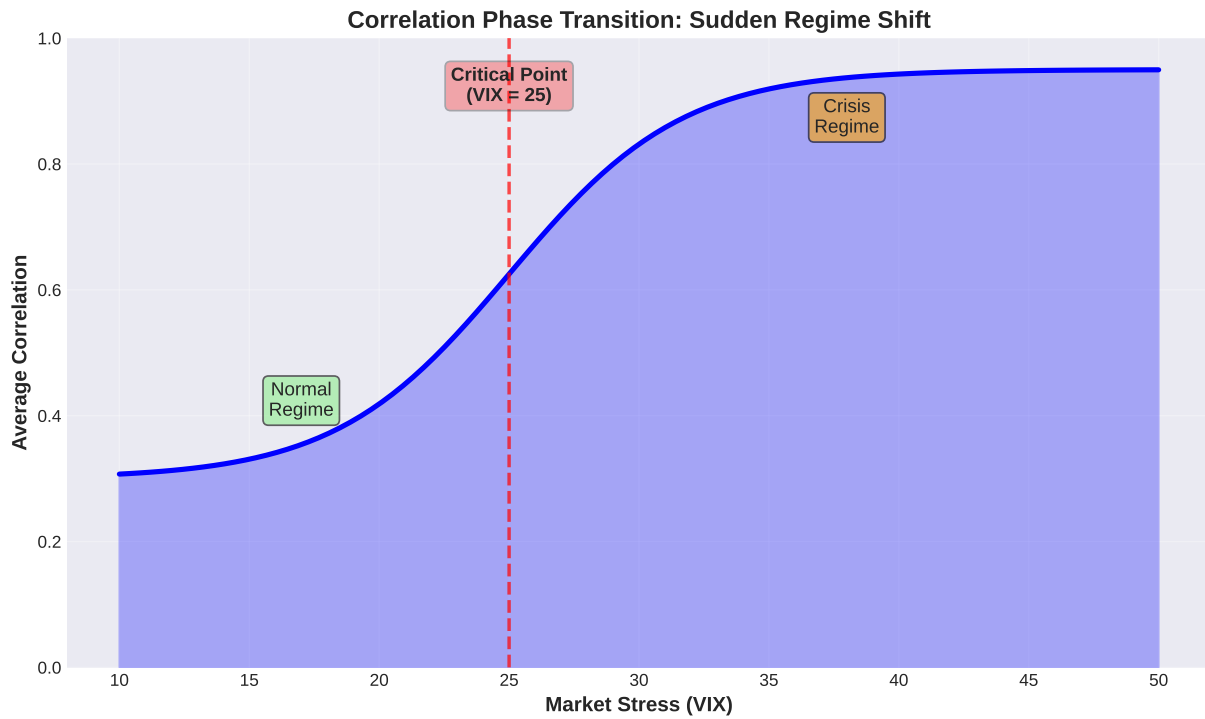


Figure 18: Correlations don't rise gradually—they snap to 1 at critical VIX level

Mathematical Model:

$$\rho(t) = \rho_{\text{base}} + \frac{\rho_{\text{max}} - \rho_{\text{base}}}{1 + e^{-\beta(\text{VIX} - \text{VIX}_{\text{crit}})}}$$

where $\text{VIX}_{\text{crit}} \approx 25$ (empirical threshold from 2008, 2020 data).

The Correlation Heatmap Trick

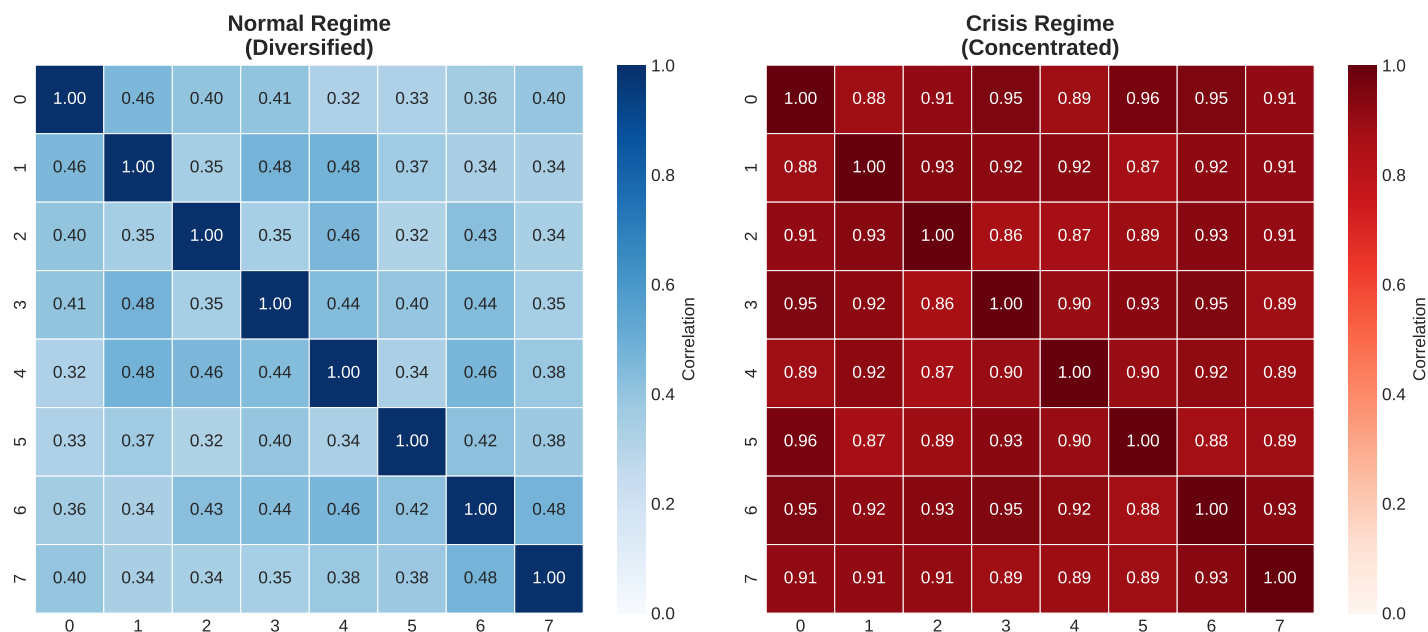


Figure 20: Visual regime detection: checkerboard = diversified, solid color = concentrated

What to Look For:

- **Checkerboard:** Low correlation (diversified)
- **Solid color:** High correlation (concentrated)
- **Block structure:** Sector clustering

The Phase Portrait (ODE Stability at a Glance)

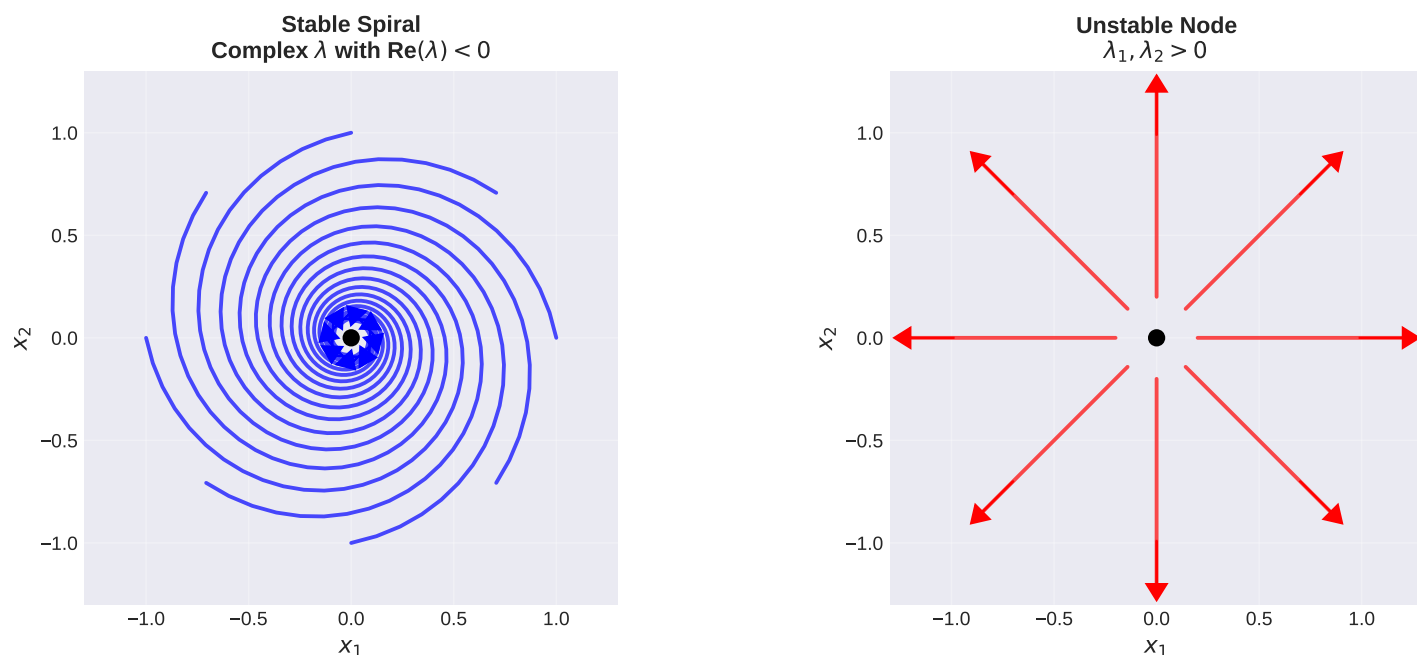


Figure 21: Arrows reveal stability instantly: inward = stable, outward = unstable